KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 10 TRANSITION

CHAPTER 2

YEAR 10 MATHEMATICS

Measurement

7 lessons · two pages each · Easy, then Medium

2.1 Converting units
2.2 Perimeter
2.3 Circumference
2.4 Area
2.5 Area of circles
2.6 Surface area
2.7 Volume and capacity

NAME _____ CLASS _____ TEACHER _____

Artwork: feathery gold spirals deep in Elephant Valley, coloured in gold and green, part of the Mandelbrot set.

How to use this booklet

1. Each lesson is **two pages**. Read the **learning intention** and tick each **success criterion** when you can do it.

2. Page 1 is **Easy** and page 2 is **Medium**. Each page starts with an example, then questions for you.

WE DO **Grey boxes.** Your teacher works each example on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Your turn. Work down the box one round at a time. Answers go on the line; show working in the boxes.

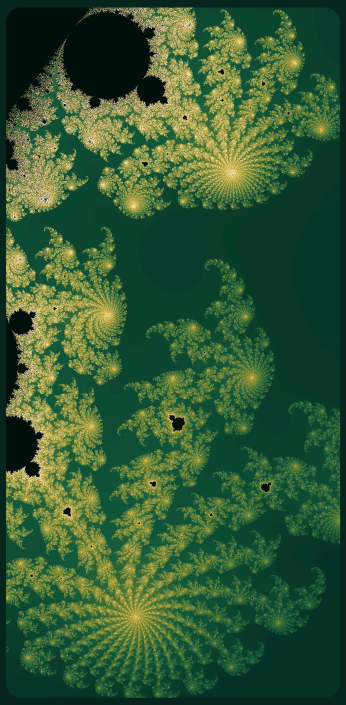
3. Stuck? Read the **Remember** line at the top of the blue box, then look back at the example you copied.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
2.1	Converting units	3		☐
2.2	Perimeter	5		☐
2.3	Circumference	7		☐
2.4	Area	9		☐
2.5	Area of circles	11		☐
2.6	Surface area	13		☐
2.7	Volume and capacity	15		☐

In this chapter you will

- convert between units of length, mass, capacity and time
- find the perimeter of shapes, including composite shapes
- find the circumference and the area of circles
- find the area of squares, rectangles, triangles, parallelograms, trapeziums and rhombuses
- find the surface area of prisms
- find the volume and capacity of prisms and cylinders



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \to z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is feathery gold spirals deep in Elephant Valley, coloured in gold and green. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON 2.1

Converting units

Date _____

Learning intention
We are learning to convert between units of length, mass, capacity and time.

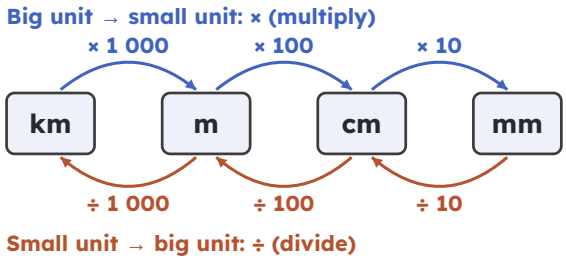
Success criteria
- ☐ I can change between km, m, cm and mm
- ☐ I can change between g, kg and t, and between mL and L
- ☐ I can change between seconds, minutes, hours and days

Level 1 · Getting started

Easy · 12 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the chart. Big unit to small unit: multiply. Small unit to big unit: divide.



a Change 4 m to cm.

b Change 7 cm to mm.

c Change 3 000 m to km.

d Change 250 cm to m.

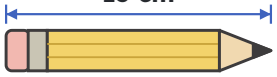
YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

12 QUESTIONS

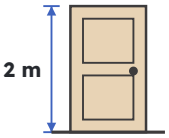
REMEMBER 1 km = 1 000 m · 1 m = 100 cm · 1 cm = 10 mm · Big unit → small unit: × · Small unit → big unit: ÷.

ROUND 1 Questions 1–4: change each length to the smaller unit. Multiply (×).

1 Change to mm.



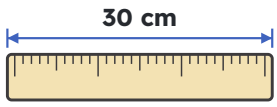
2 Change to cm.



3 Change to m.



4 Change to mm.

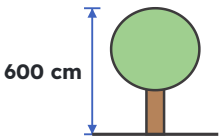


ROUND 2 Questions 5–8: change each length to the bigger unit. Divide (÷).

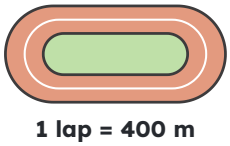
5 Change to cm.



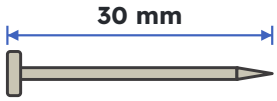
6 Change to m.



7 Change to km.

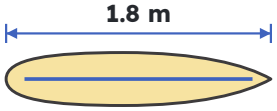


8 Change to cm.

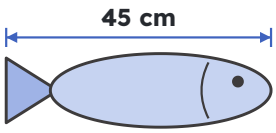


ROUND 3 Questions 9–12: multiply or divide? Change each length to the unit shown.

9 Change to cm.



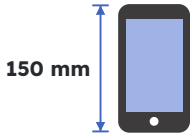
10 Change to m.



11 Change to m.



12 Change to cm.



2.1 Converting units

MEDIUM

Level 2 · Building up

Medium · 14 questions

WE DO With your teacher: copy what they write on the board.

Example 2 **Mass and capacity use × 1 000. Time uses × 60 and × 24.**

Mass

t ⇄ (× 1 000 / ÷ 1 000) kg ⇄ (× 1 000 / ÷ 1 000) g

Capacity

L ⇄ (× 1 000 / ÷ 1 000) mL

Time

day ⇄ (× 24 / ÷ 24) h ⇄ (× 60 / ÷ 60) min ⇄ (× 60 / ÷ 60) s

t = tonnes · h = hours · min = minutes · s = seconds

a Change 3 kg to g.

b Change 2 500 mL to L.

c Change 5 minutes to seconds.

d Change 2 hours to seconds.

YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

14 QUESTIONS

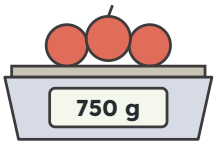
REMEMBER 1 t = 1 000 kg · 1 kg = 1 000 g · 1 L = 1 000 mL · 1 day = 24 h · 1 h = 60 min · 1 min = 60 s.

ROUND 1 Questions 1–4: mass. Change each one to the unit shown.

1 Change to g.



2 Change to kg.



3 Change to kg.



4 Change to tonnes.



ROUND 2 Questions 5–8: capacity. Change each one to the unit shown.

5 Change to mL.



6 Change to L.



7 Change to L.



8 Change to mL.



ROUND 3 Questions 9–12: time. Change each one to the unit shown.

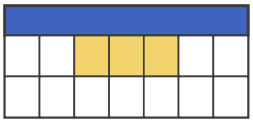
9 Change to seconds.



10 Change to minutes.



11 Change to hours.



12 Change to seconds.



ROUND 4 Questions 13–14: solve each problem. Show your working.

13 Ava is in a 5.3 km fun run. Ava has run 2 800 m so far. How many metres does Ava still have to run?

14 A jug holds 2 L of juice. Each glass holds 250 mL. How many glasses can Leo fill?

LESSON 2.2

Perimeter

Date _____

Learning intention
We are learning to find the perimeter of a shape by adding up the lengths of all its sides.

Success criteria
- ☐ I can add up the sides of a shape to find its perimeter
- ☐ I can use tick marks to find sides that are the same length
- ☐ I can find missing sides first, then the perimeter

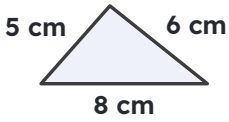
Level 1 · Getting started

Easy · 12 questions

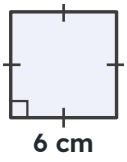
WE DO With your teacher: copy what they write on the board.

Example 1 Perimeter is the distance around the outside. Add up the lengths of all the sides.

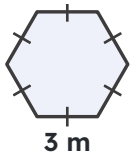
a Triangle



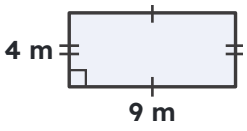
b Square



c Hexagon



d Rectangle



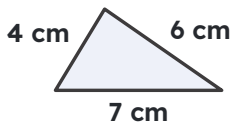
YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

12 QUESTIONS

REMEMBER Perimeter (P) = add up the lengths of all the sides · Sides with the same tick marks are the same length · Square: P = 4 × side.

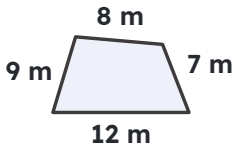
ROUND 1 Questions 1–4: find the perimeter of each shape. Add up all the sides.

1



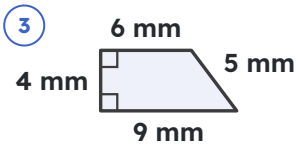
P = _____

2



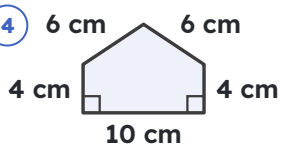
P = _____

3



P = _____

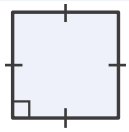
4



P = _____

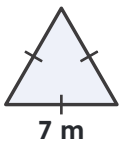
ROUND 2 Questions 5–8: every side is the same length (the tick marks show this). Find each perimeter.

5



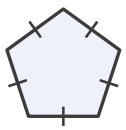
P = _____

6



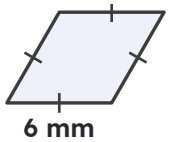
P = _____

7



P = _____

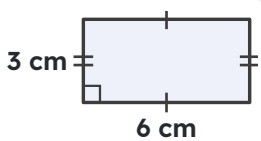
8



P = _____

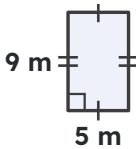
ROUND 3 Questions 9–12: sides with the same tick marks are the same length. Find each perimeter.

9



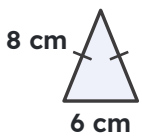
P = _____

10



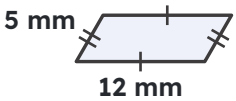
P = _____

11



P = _____

12



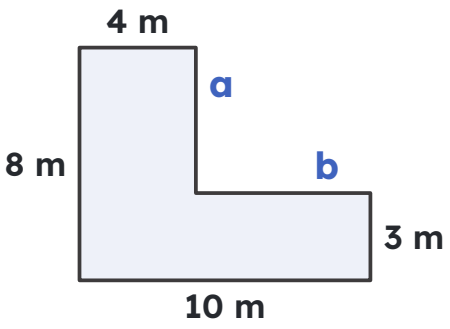
P = _____

Level 2 · Building up

Medium · 13 questions

WE DO With your teacher: copy what they write on the board.

Example 2 **Sides a and b have no length on them. Find them first, then add up all the sides.**



a Find side a.

b Find side b.

c Find the perimeter.

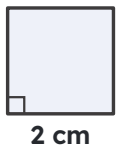
d Fencing costs $20 a metre. Find the cost to fence the shape.

YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes). 13 QUESTIONS

REMEMBER Opposite sides of a rectangle are the same length · A missing side: the whole side = the two parts added, so take away the part you know · Then add up every side.

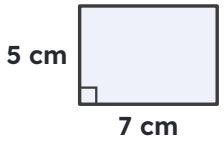
ROUND 1 Questions 1–4: find each perimeter. Use the sides you know to find the others.

1 Square



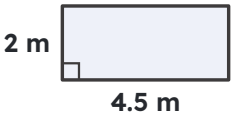
P = _____

2 Rectangle



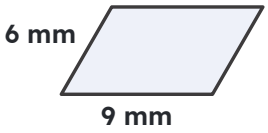
P = _____

3 Rectangle



P = _____

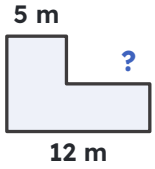
4 Parallelogram



P = _____

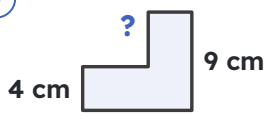
ROUND 2 Questions 5–8: find the length of each side marked ?.

5



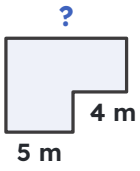
? = _____

6



? = _____

7



? = _____

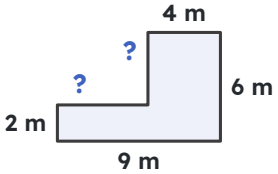
8



? = _____

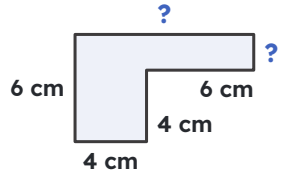
ROUND 3 Questions 9–11: find the perimeter of each shape. Find the ? sides first.

9



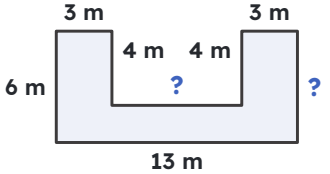
P = _____

10



P = _____

11



P = _____

ROUND 4 Questions 12–13: solve each problem. Show your working.

12 A rectangular paddock is 50 m long and 30 m wide. Fencing costs $12 a metre. How much does it cost to fence the paddock?

13 Zoe runs around a square park 3 times. Each side of the park is 200 m long. How far does Zoe run?

LESSON 2.3

Circumference

Date _____

Learning intention
We are learning to find the circumference of a circle, and the perimeter of semicircles and quarter circles.

Success criteria
- ☐ I can change a radius to a diameter, and back
- ☐ I can use $C = \pi \times d$ and $C = 2 \times \pi \times r$
- ☐ I can find the perimeter of shapes with a curved side

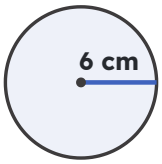
Level 1 · Getting started

Easy · 12 questions

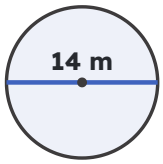
WE DO With your teacher: copy what they write on the board.

Example 1 The circumference (C) is the distance around a circle. Round to 1 decimal place.

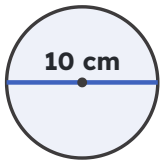
a Find the diameter.



b Find the radius.



c Find C.



d Find C.



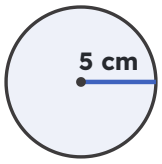
YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

12 QUESTIONS

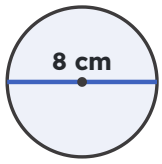
REMEMBER Radius r: from the centre to the edge · Diameter d: all the way across, $d = 2 \times r$ · $C = \pi \times d$ or $C = 2 \times \pi \times r$. Use the π button, then round to 1 decimal place.

ROUND 1 Questions 1–4: find the diameter ($d = 2 \times r$) or the radius ($r = d \div 2$).

1 Find the diameter.



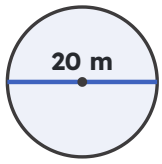
2 Find the radius.



3 Find the diameter.



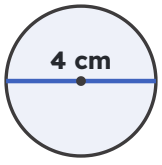
4 Find the radius.



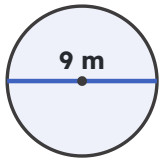
_____ _____ _____ _____

ROUND 2 Questions 5–8: the diameter is shown. Find each circumference: $C = \pi \times d$.

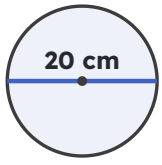
5



6



7



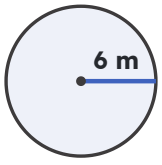
8



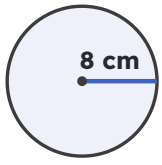
C = _____ C = _____ C = _____ C = _____

ROUND 3 Questions 9–12: the radius is shown. Find each circumference: $C = 2 \times \pi \times r$.

9



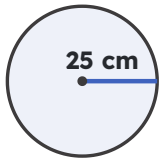
10



11



12



C = _____ C = _____ C = _____ C = _____

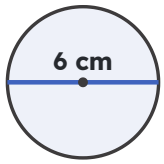
Level 2 · Building up

Medium · 15 questions

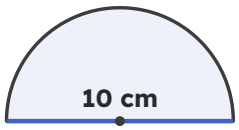
WE DO With your teacher: copy what they write on the board.

Example 2 **Perimeter = the curved part + the straight sides. Round to 1 decimal place.**

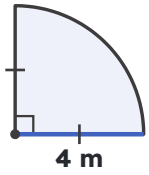
a Circle



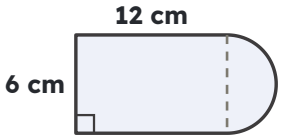
b Semicircle



c Quarter circle



d Composite shape



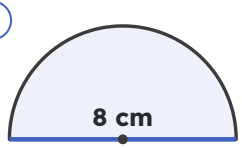
YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

15 QUESTIONS

REMEMBER Semicircle: P = ½ × π × d + d · Quarter circle: P = ¼ × 2 × π × r + r + r · Composite shape: add the straight sides and the curved part (not the dashed line).

ROUND 1 Questions 1–4: find the perimeter of each semicircle. If the radius is shown, find d first.

1



P = _____

2



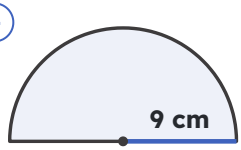
P = _____

3



P = _____

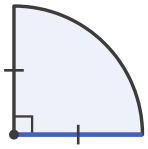
4



P = _____

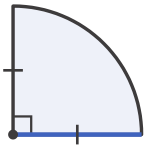
ROUND 2 Questions 5–8: find the perimeter of each quarter circle.

5



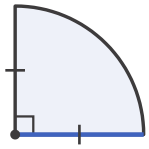
P = _____

6



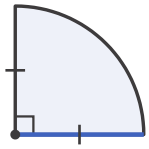
P = _____

7



P = _____

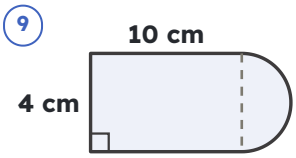
8



P = _____

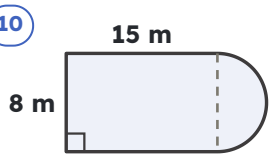
ROUND 3 Questions 9–12: find the perimeter of each shape. Do not count the dashed line.

9



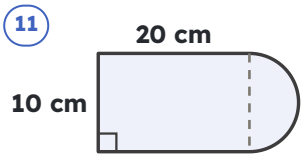
P = _____

10



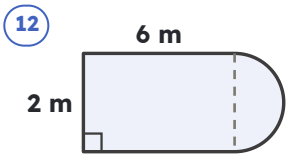
P = _____

11



P = _____

12



P = _____

ROUND 4 Questions 13–15: solve each problem. Round to 1 decimal place.

13 A round garden bed has a radius of 3 m. Edging goes all the way around it. How much edging is needed?

14 A bike wheel has a diameter of 60 cm. How far does the bike move when the wheel turns once?

15 A window is a semicircle with a diameter of 2 m. Lights go all the way around its edge. How long are the lights?

LESSON 2.4

Area

Date _____

Learning intention
We are learning to find the area of common shapes, from rectangles to trapeziums.

Success criteria
- ☐ I can find the area of squares, rectangles and triangles
- ☐ I can find the area of parallelograms, trapeziums and rhombuses
- ☐ I can write area in square units (cm², m²)

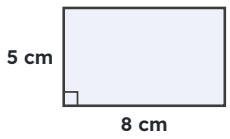
Level 1 · Getting started

Easy · 12 questions

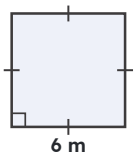
WE DO With your teacher: copy what they write on the board.

Example 1 Find the area of each shape. Area is measured in square units (cm², m²).

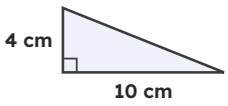
a Rectangle



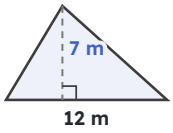
b Square



c Right-angled triangle



d Triangle



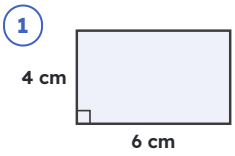
YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

12 QUESTIONS

REMEMBER Rectangle: A = length × width · Square: A = side × side · Triangle: A = ½ × base × height (the height meets the base at a right angle).

ROUND 1 Questions 1–4: find the area of each rectangle and square.

1



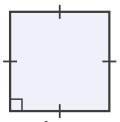
A = _____

2



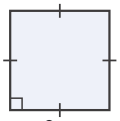
A = _____

3



A = _____

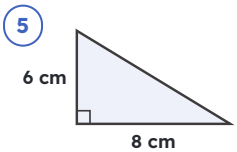
4



A = _____

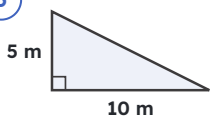
ROUND 2 Questions 5–8: find the area of each right-angled triangle.

5



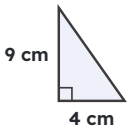
A = _____

6



A = _____

7



A = _____

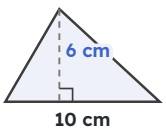
8



A = _____

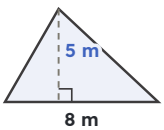
ROUND 3 Questions 9–12: find the area of each triangle. Use the dashed height.

9



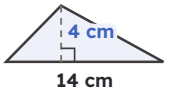
A = _____

10



A = _____

11



A = _____

12



A = _____

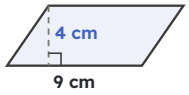
Level 2 · Building up

Medium · 14 questions

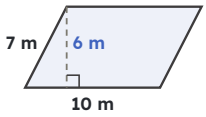
WE DO With your teacher: copy what they write on the board.

Example 2 Use the perpendicular height (the dashed line), not the slanted side.

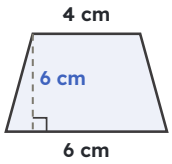
a Parallelogram



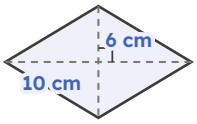
b Parallelogram



c Trapezium



d Rhombus



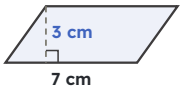
YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

14 QUESTIONS

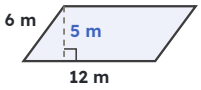
REMEMBER Parallelogram: A = base × height · Trapezium: A = ½ × (a + b) × h · Rhombus: A = ½ × x × y (x and y are the diagonals).

ROUND 1 Questions 1–4: find the area of each parallelogram.

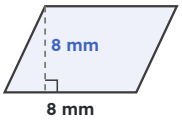
1



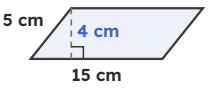
2



3



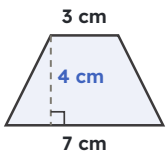
4



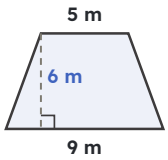
A = _____ A = _____ A = _____ A = _____

ROUND 2 Questions 5–8: find the area of each trapezium.

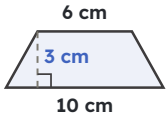
5



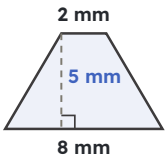
6



7



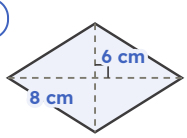
8



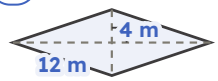
A = _____ A = _____ A = _____ A = _____

ROUND 3 Questions 9–12: find the area of each rhombus.

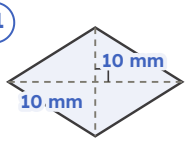
9



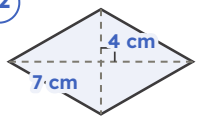
10



11



12



A = _____ A = _____ A = _____ A = _____

ROUND 4 Questions 13–14: solve each problem. Show your working.

13 A rectangular lawn is 12 m long and 8 m wide. Turf costs $15 for each m². How much does it cost to turf the lawn?

14 A triangular sail has a base of 3 m and a height of 5 m. Find the area of the sail.

LESSON 2.5

Area of circles

Date _____

Learning intention
We are learning to find the area of circles, semicircles and quarter circles.

Success criteria
- ☐ I can find the area of a circle from its radius
- ☐ I can halve the diameter to find the radius first
- ☐ I can find the area of semicircles, quarter circles and composite shapes

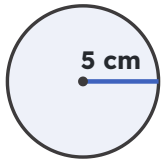
Level 1 · Getting started

Easy · 12 questions

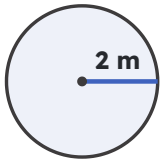
WE DO With your teacher: copy what they write on the board.

Example 1 $A = \pi \times r^2$. If the diameter is shown, halve it to get r. Round to 1 decimal place.

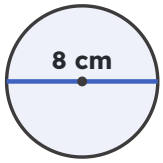
a Radius shown



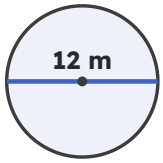
b Radius shown



c Diameter shown



d Diameter shown



YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

12 QUESTIONS

REMEMBER $A = \pi \times r^2$ ($r^2 = r \times r$) · Diameter shown? Halve it first to get r · Use the π button, then round to 1 decimal place.

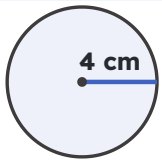
ROUND 1 Questions 1–4: the radius is shown. Find each area.

1



A = _____

2



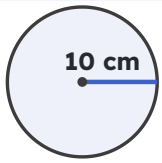
A = _____

3



A = _____

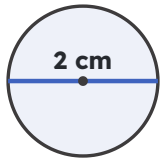
4



A = _____

ROUND 2 Questions 5–8: the diameter is shown. Halve it, then find each area.

5



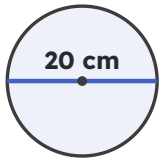
A = _____

6



A = _____

7



A = _____

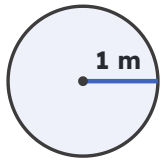
8



A = _____

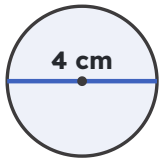
ROUND 3 Questions 9–12: radius or diameter? Find each area.

9



A = _____

10



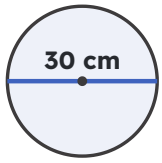
A = _____

11



A = _____

12



A = _____

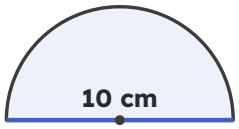
Level 2 · Building up

Medium · 15 questions

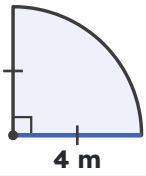
WE DO With your teacher: copy what they write on the board.

Example 2 **A semicircle is half a circle; a quarter circle (a 90° sector) is a quarter. Round to 1 decimal place.**

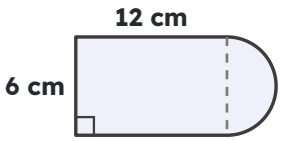
a Semicircle



b Quarter circle



c Composite shape



d Semicircle



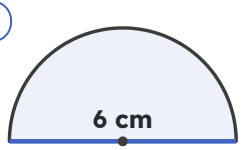
YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

15 QUESTIONS

REMEMBER Semicircle: $A = \frac{1}{2} \times \pi \times r^2$ · Quarter circle: $A = \frac{1}{4} \times \pi \times r^2$ · Composite shape: add the area of each part.

ROUND 1 Questions 1–4: find the area of each semicircle. If the diameter is shown, halve it first.

1



A = _____

2



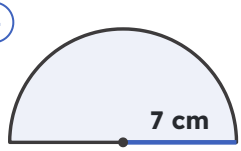
A = _____

3



A = _____

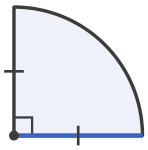
4



A = _____

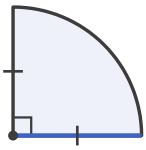
ROUND 2 Questions 5–8: find the area of each quarter circle.

5



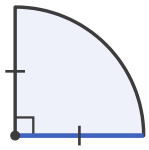
A = _____

6



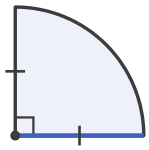
A = _____

7



A = _____

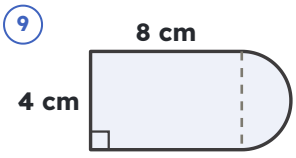
8



A = _____

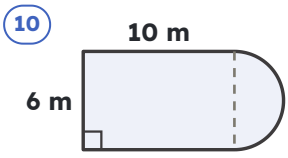
ROUND 3 Questions 9–12: find the area of each shape: the rectangle + the semicircle.

9



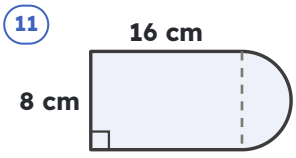
A = _____

10



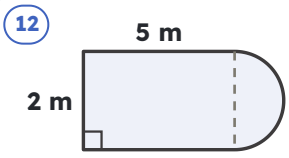
A = _____

11



A = _____

12



A = _____

ROUND 4 Questions 13–15: solve each problem. Round to 1 decimal place.

13 A round pizza has a diameter of 30 cm. What is the area of the pizza?

14 A circular rug has a radius of 1.5 m. What area of floor does it cover?

15 A sprinkler waters a circle with a radius of 4 m. What area of lawn does it water?

LESSON 2.6

Surface area

Date _____

Learning intention
We are learning to find the surface area of a prism by adding the areas of all its faces.

Success criteria
- ☐ I can find the surface area of a cube
- ☐ I can find the surface area of a rectangular prism
- ☐ I can find the surface area of a triangular prism

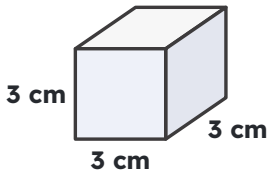
Level 1 · Getting started

Easy · 12 questions

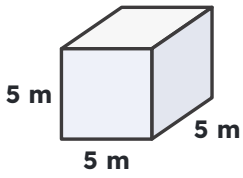
WE DO With your teacher: copy what they write on the board.

Example 1 **Surface area (SA) = the total area of all the faces. A rectangular prism has 6 faces in 3 matching pairs.**

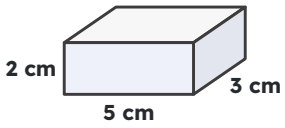
a Cube



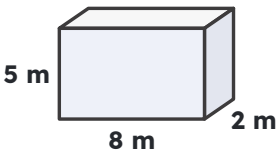
b Cube



c Rectangular prism



d Rectangular prism



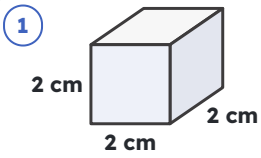
YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

12 QUESTIONS

REMEMBER SA = 2 × top + 2 × front + 2 × side (top = length × width, front = length × height, side = width × height) · Cube: SA = 6 × side × side.

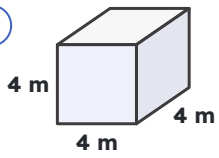
ROUND 1 Questions 1–4: find the surface area of each cube.

1



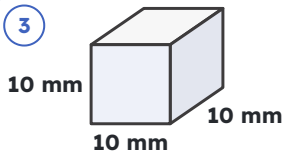
SA = _____

2



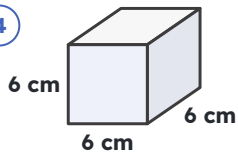
SA = _____

3



SA = _____

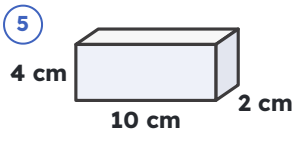
4



SA = _____

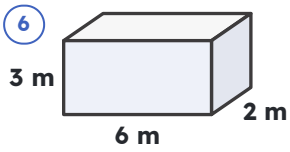
ROUND 2 Questions 5–8: find the surface area of each rectangular prism.

5



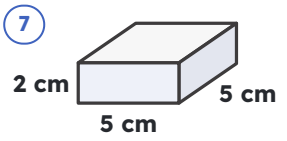
SA = _____

6



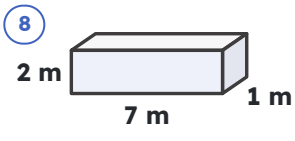
SA = _____

7



SA = _____

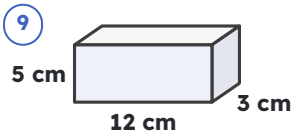
8



SA = _____

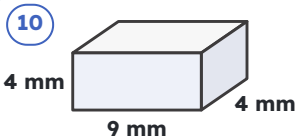
ROUND 3 Questions 9–12: find the surface area of each rectangular prism.

9



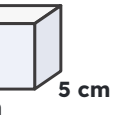
SA = _____

10



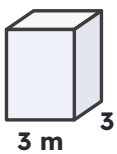
SA = _____

11



SA = _____

12



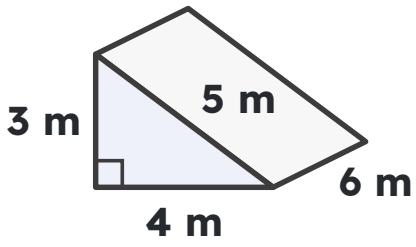
SA = _____

Level 2 · Building up

Medium · 14 questions

WE DO With your teacher: copy what they write on the board.

Example 2 **A triangular prism has 5 faces: 2 triangles and 3 rectangles. Add them all.**



a How many faces does a triangular prism have?

b Find the area of one triangle.

c Find the area of the three rectangles.

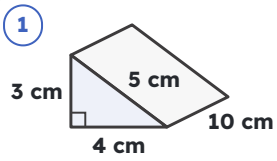
d Find the surface area.

YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes). 14 QUESTIONS

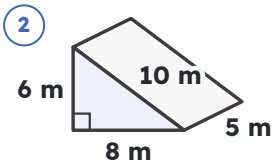
REMEMBER Each triangle: ½ × base × height · Each rectangle: the prism's length × one side of the triangle · SA = 2 triangles + 3 rectangles.

ROUND 1 Questions 1–4: find the surface area of each triangular prism (right-angled ends).

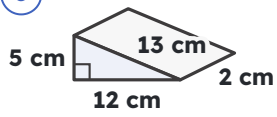
1



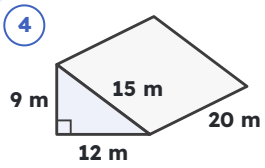
2



3



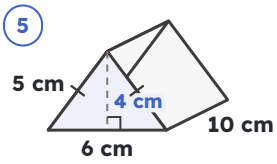
4



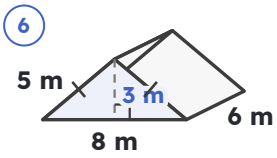
SA = _____ SA = _____ SA = _____ SA = _____

ROUND 2 Questions 5–8: find the surface area of each triangular prism. The dashed line is the height.

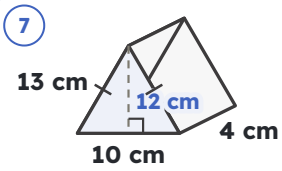
5



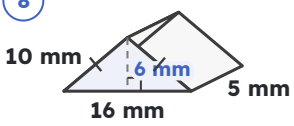
6



7



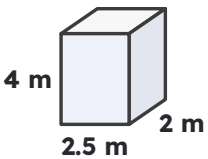
8



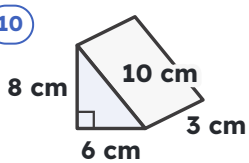
SA = _____ SA = _____ SA = _____ SA = _____

ROUND 3 Questions 9–12: rectangular or triangular prism? Find each surface area.

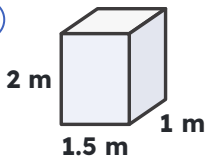
9



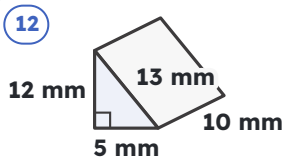
10



11



12



SA = _____ SA = _____ SA = _____ SA = _____

ROUND 4 Questions 13–14: solve each problem. Show your working.

13 A gift box is a cube with sides of 20 cm. How much wrapping paper covers the whole box (with no overlap)?

14 A wooden box is 50 cm long, 30 cm wide and 20 cm high. Every face is painted. What area is painted?

LESSON 2.7

Volume and capacity

Date _____

Learning intention
We are learning to find the volume of prisms and cylinders, and change volume to capacity.

Success criteria
- ☐ I can find the volume of a cube and a rectangular prism
- ☐ I can find the volume of a triangular prism and a cylinder
- ☐ I can change cm^3 to mL and L

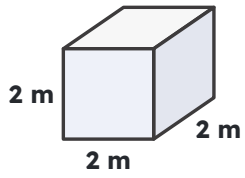
Level 1 · Getting started

Easy · 12 questions

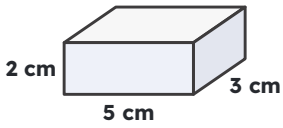
WE DO With your teacher: copy what they write on the board.

Example 1 **Volume (V) is the space inside a solid. V = length × width × height. Volume is in cubic units (cm^3, m^3).**

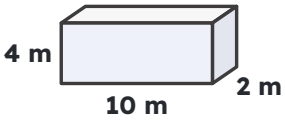
a Cube



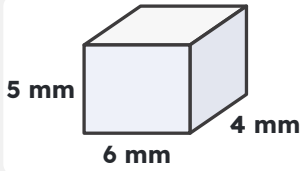
b Rectangular prism



c Rectangular prism



d Rectangular prism



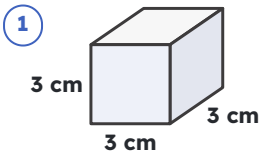
YOU DO On your own: **Easy** work down the box, one round at a time. Write each answer on its line.

12 QUESTIONS

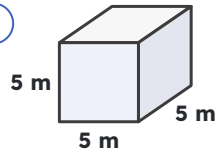
REMEMBER V = length × width × height · Cube: V = side × side × side · Volume units are cubed (cm^3, m^3).

ROUND 1 Questions 1–4: find the volume of each cube.

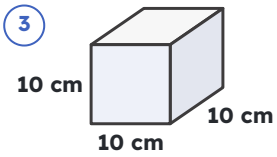
1.



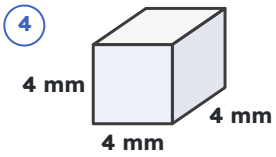
2.



3.



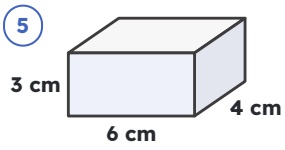
4.



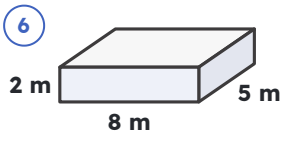
V = _____ V = _____ V = _____ V = _____

ROUND 2 Questions 5–8: find the volume of each rectangular prism.

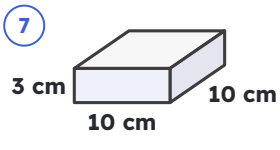
5.



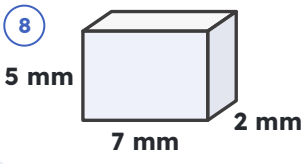
6.



7.



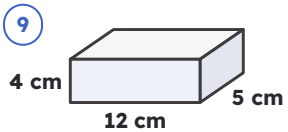
8.



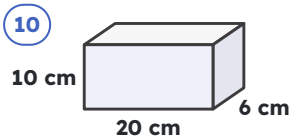
V = _____ V = _____ V = _____ V = _____

ROUND 3 Questions 9–12: find the volume of each rectangular prism.

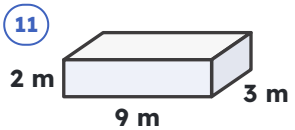
9.



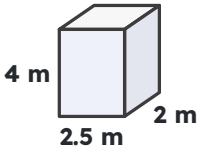
10.



11.



12.



V = _____ V = _____ V = _____ V = _____

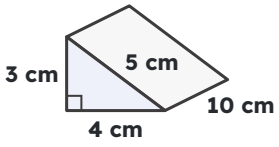
Level 2 · Building up

Medium · 14 questions

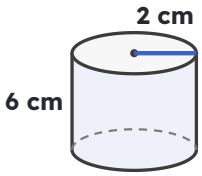
WE DO With your teacher: copy what they write on the board.

Example 2 **Prism: V = area of the end × length · Cylinder: V = π × r² × h · 1 cm³ holds 1 mL.**

a Triangular prism



b Cylinder



c How many millilitres does the cylinder in b hold?

d A box holds 2 000 cm³. How many litres is that?

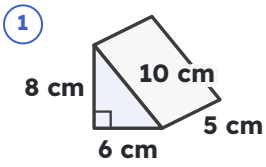
YOU DO On your own: **Medium** work down the box, one round at a time. Write each answer on its line (show working in the boxes).

14 QUESTIONS

REMEMBER Triangular prism: V = ½ × base × height × length · Cylinder: V = π × r² × h (round to 1 d.p.) · 1 cm³ = 1 mL · 1 000 cm³ = 1 L · 1 m³ = 1 000 L.

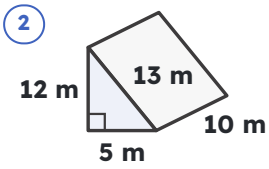
ROUND 1 Questions 1–4: find the volume of each triangular prism.

1



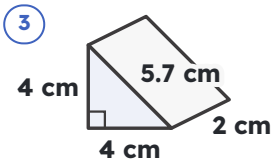
V = _____

2



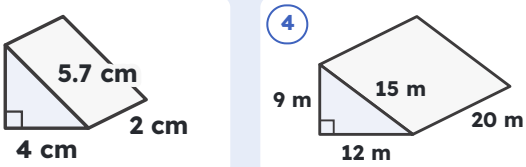
V = _____

3



V = _____

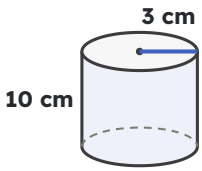
4



V = _____

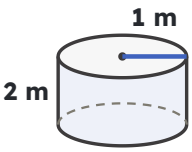
ROUND 2 Questions 5–8: find the volume of each cylinder. Round to 1 decimal place.

5



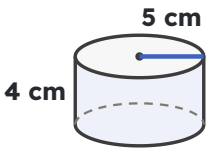
V = _____

6



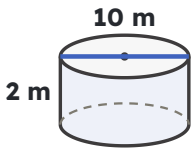
V = _____

7



V = _____

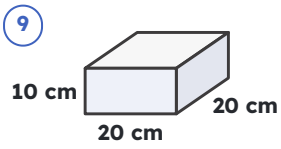
8



V = _____

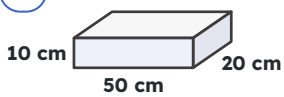
ROUND 3 Questions 9–12: find the volume of each box in cm³, then its capacity in litres (÷ 1 000).

9



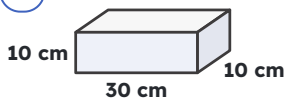
L = _____

10



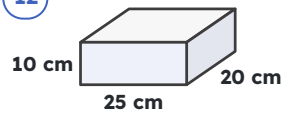
L = _____

11



L = _____

12



L = _____

ROUND 4 Questions 13–14: solve each problem. Show your working.

13 A water tank is a cylinder with a diameter of 8 m and a height of 2 m. Find its volume (to 1 decimal place). How many litres does it hold? (1 m³ = 1 000 L)

14 A fish tank is 60 cm long, 30 cm wide and 40 cm high. How many litres of water does it hold when full?